# EESSI

EUROPEAN ENVIRONMENT FOR SCIENTIFIC SOFTWARE INSTALLATIONS

5 Mar 2026

https://github.com/EESSI/meetings/wiki

# Agenda

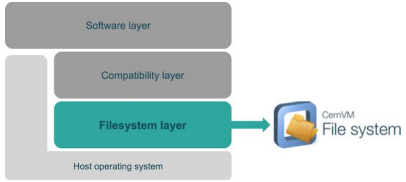
- Quick introduction by new people
- Progress update per EESSI layer
- Update on Spack integration
- Update on EESSI production repository `software.eessi.io` (+ RISC-V)
- Update on EESSI test suite, build-and-deploy bot, documentation, etc.
- AWS sponsorship update
- EESSI governance
- Upcoming/recent events
- Q&A

# Quick introduction by new people

**New people on the call: feel free to introduce yourself!**

- Who are you, where do you work, on what?
- Why are you interested in the EESSI project?
- Are you planning to actively contribute,
  and if so, to which aspect(s) of the project?

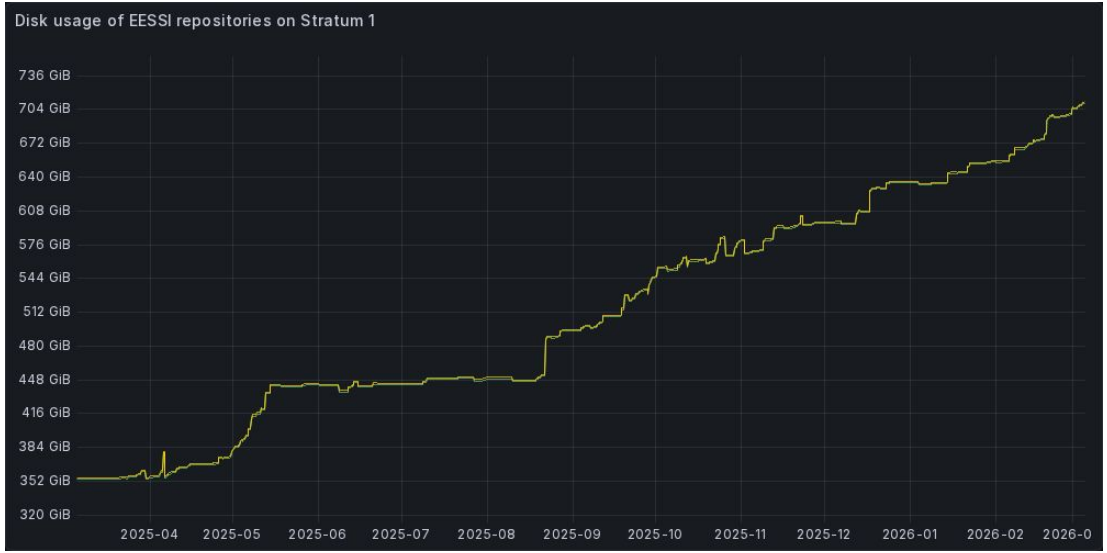
# Progress update: filesystem layer




- Add new host injection variant symlinks for 2025.06 (PR #263)
  - Used for making host GPU libraries available, see later slide
- No issues with the CernVM-FS infrastructure
- TODO: new release of package to configure CernVM-FS
  - Including info on recently added Stratum-1 mirror server hosted by RAL (UK)

[Bob, Kenneth]

# Progress update: filesystem layer monitoring

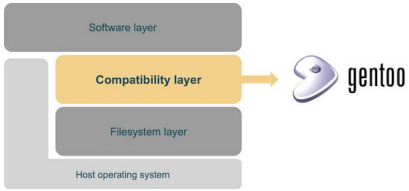



Size is growing with ~1GB/day

Currently ~710GB of disk space is required
for a full Stratum-1 mirror of all EESSI repositories

[Bob, Kenneth]

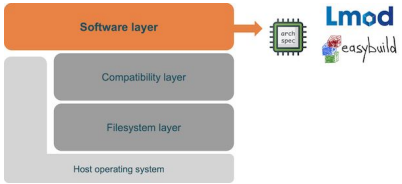
# Progress update: compatibility layer




- Glibc has been recompiled to support new host injection variant symlinks
  - See issue #226 and PR #227
  - See later slide
- The environment modules package has been installed
  - Also see https://gitlab.com/eessi/support/-/issues/195
- Open issue in EESSI support portal to collect ideas for the 2026.x compatibility layer: https://gitlab.com/eessi/support/-/issues/220

[Bob, Kenneth]

# Progress update: software layer (1/2)




Highlights of recently merged software PRs (2023.06):

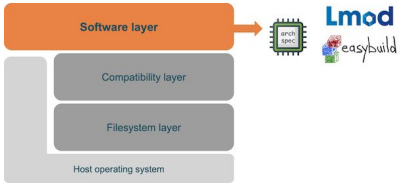
- 655 distinct software packages available now,
  10 merged PRs since last update meeting (15 Jan'26)

Highlights of recently merged software PRs (2025.06):

- 393 distinct software packages now available,
  57 merged PRs since last update meeting (15 Jan'26)
- Zen 5 builds are now available, rapidly reaching parity with other targets (see later slide)
- CUDA 12.6.0,12.8.0, cuDNN 9.5.0.50,9.10.1.4 (PR #1410) (see later slide)
- LLVM toolchain lfoss/2025b is available (PR #1360)
- More top level applications available including desktop environments / visualization modules:
  Xfce 4.20 (PR #1398), Wayland 1.4.0 (PR #1382)

[Kenneth, Thomas, Richard, Pedro, Bob, Lara, Caspar, Satish, Alan, Mattias, Danilo, Helena …]

# Progress update: software layer (2/2)

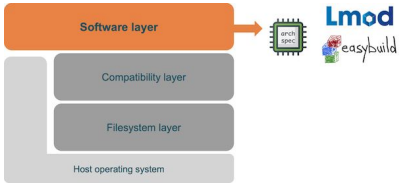



Rebuilds of existing software installations:

- EESSI 2023.06:
  - hatchling 1.18.0 (PR #1376), jupyter-server (PR #1366)
- EESSI 2025.06
  - FFmpeg 7.0.2 + 7.1.1 (PR #1422)
  - Rebuild GST-plugins-bad v1.26.6 with more plugins (PR #1404)
  - Pango 1.57.0 rebuild (PR #1387 + PR #1390)
  - Rebuild Python 3.13.5 with ctypes patch (PR #1364)
  - Rebuild libarchive v3.8.1 (PR #1383)

[Kenneth, Thomas, Richard, Pedro, Bob, Lara, Caspar, Satish, Alan, Helena …]

# Progress update: software-layer-scripts



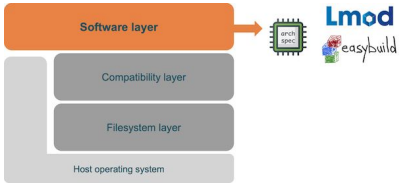

Highlights of recently merged software-layer-scripts PRs:

- Add CUDA 12.9.1 and cuDNN-9.15.0.57 support to EESSI 2025.06 (PR #167)
- Make all Lmod-based shell initialisations more robust, add additional features to facilitate site integration (PR #153)
- let EESSI container script's `run` mode call `apptainer exec` instead of `apptainer run` (PR #152)

+ various other minor PRs: #148, #149, #151, #152, #154, #158, #159, #160, #161, #164, #165, #166, #168, #169

[Kenneth, Alan, Bob, Richard, Thomas …]

# Progress update: RISC-V




RISC-V software stack being built via https://github.com/EESSI/dev.eessi.io-riscv

- Installations targeting `/cvmfs/dev.eessi.io/riscv` repo
  - Same installation workflow through bot
  - Built on top of EESSI 2025.06 compatibility layer for RISC-V (available in `software.eessi.io`)
- >175 installations have already been ingested 🎉
  - `foss/2025a` and `foss/2025b` are available
- Planning to also look into:
  - More end-user applications
  - LLVM-based toolchains (`lfoss`)
  - Providing an overview of available software in the documentation
  - Going beyond `riscv64/generic` as target, also optimize for Banana Pi (RISC-V CPU w/ vector instructions)

[Bob, Julian, Kenneth, …]

# Using Spack to install software on top of EESSI

- Update on status in blog post: eessi.io/docs/blog/2026/02/05/Spack-on-top-of-EESSI-best-of-...
- Spack 1.1+ introduces external packages treated as concrete specs
  - Allowing us to expose software installed in EESSI as externals along with their dependencies
  - `spack external find` to discover dependencies in the compatibility layer
    - Helps, but not perfect
- Able to build a new Quantum ESPRESSO with Spack including MPI support (+ all EESSI deps)
- Demonstrated implementation now available in the Spood repository
- Need others to carry on the work!

[Loris, Alan, Kenneth, …]

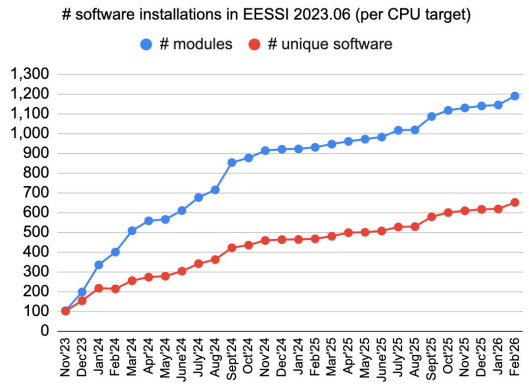
# EESSI production repository

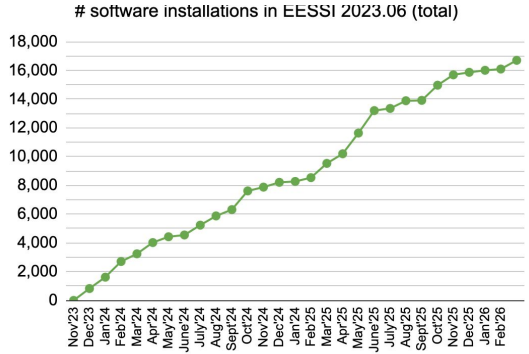
eessi.io/docs

Status update on **production-ready EESSI repository** `software.eessi.io`

Version **2023.06** is being populated with software via PRs to software-layer repo + build-and-deploy bot

- **14** fully supported CPU targets: see eessi.io/docs/software_layer/cpu_targets
- Support for NVIDIA GPUs is in place, see eessi.io/docs/gpu
- **Currently: 1,189 software installations per CPU target**
  (+60 in Jan+Feb'26)
  - **651 different software projects** (+35 in Jan+Feb'26)
    16,709 software installations (across 14 CPU targets, +700 in Jan+Feb'26)
- Current focus:
  - Adding more GPU (CUDA) software installations
  - Processing incoming contributions
  - Software layer for new EESSI version 2025.06 (see next slide)



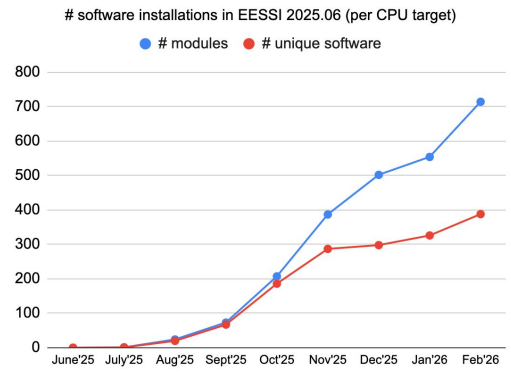
# EESSI production repository

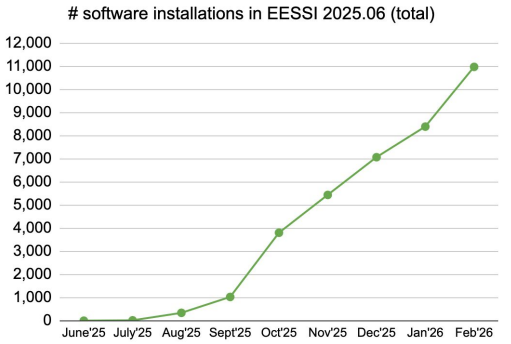
eessi.io/docs

Status update on **production-ready EESSI repository** `software.eessi.io`

Version **2025.06** is being populated with software via PRs to software-layer repo + build-and-deploy bot

- 14 fully supported CPU targets: see eessi.io/docs/software_layer/cpu_targets
- ⏳ **Extra CPU target is being added: AMD Zen 5 (Turin)**
- Support for NVIDIA GPUs is in place, see eessi.io/docs/gpu
- **Currently: 714 software installations per CPU target** (+220 in Jan+Feb'26)
  - **388 different software projects** (+90 in Jan+Feb'26)
    10,967 software installations (across 14 CPU targets, +3,903 in Jan+Feb'26)
- ✅ LLVM-based `lfoss/2025b` toolchain was added
- ✅ Overview of available software in documentation also covers EESSI 2025.06
- Current focus:
  - Add CUDA installations
  - Add support for Zen 5 (and soon also Graviton 4+5)
  - Processing incoming contributions



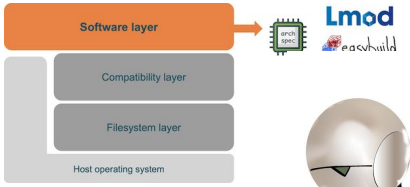



# Support for AMD Zen 5 (Turin)

- Work in progress to add `x86_64/amd/zen5` to EESSI 2025.06, >80% of the software has been built
  - Using scripts from https://github.com/EESSI/software-layer-scripts/pull/163 to reproduce the zen2 stack
  - No big issues encountered: https://github.com/EESSI/software-layer/issues/1430
  - Accelerator stacks (CUDA + cuDNN) still have to be built
- Draft PRs to add Zen 5 support to CPU autodetection and CI workflows (PR #1395)
- Still has to be tested
  - Quite a lot of differences in reported CPU flags between different Zen 5 nodes
  - Does this have impact on performance?
    - Benchmark EESSI stack (built on AWS VM) vs native stack
- Try it out:

```
export EESSI_SOFTWARE_SUBDIR_OVERRIDE=x86_64/amd/zen5
source /cvmfs/software.eessi.io/versions/2025.06/init/lmod/bash
```

# Bot for building + deploying software



https://github.com/EESSI/eessi-bot-software-layer + https://github.com/EESSI/filesystem-layer

Bot version 0.11.0 has been released (28 Jan'26) release notes

- Bug fixes
  - solves issue that not all build jobs were considered for `bot: status`
- Improvements:
  - support for new command `bot: cancel jobid:<JOBID> ...`
- Changes to bot configuration (`app.cfg`)
  - value for `no_build_permission_comment` was updated
  - NEW `cancel_command` -> path to command for cancelling jobs (e.g., scancel)
  - NEW `api_timeout` -> timeout for querying comments via GitHub REST API

[Bob, Caspar, Kenneth, Lara, Pedro, Sam, Sondre, Thomas]

# EESSI documentation - available software update

eessi.io/docs/available_software/

- Now leverages data from https://www.eessi.io/api_data/
  - Includes both EESSI versions
  - GPU software metadata
- Much improved views
  - Includes software/extension count
  - Uses cards
  - Better searchability
  - Improved cpu/gpu arch overviews

## Software available in EESSI

Overview of software available in EESSI's production repository `software.eessi.io`.

*740 unique software projects (+ 2995 unique extensions)*

gromacs ×

**AITW-viscosity** *(more details)*

https://www.ai-transpwood-project.eu

AI-TranspWood - AiiDA workchain to compute the shear viscosity of molecular liquids, starting from a SMILES string of a molecule and using GROMACS to perform the molecular dynamics simulations.

*Available in EESSI versions:* 2023.06

*Supported CPU families:* AMD Intel Arm

*Supported GPU families: (none)*

**GROMACS** *(more details)*

https://www.gromacs.org

GROMACS is a versatile package to perform molecular dynamics, i.e. simulate the Newtonian equations of motion for systems with hundreds to millions of particles. This is a CPU only build, containing both MPI and threadMPI binaries for both single and double precision. It also contains the gmxapi extension for the single precision MPI build.

*Available in EESSI versions:* 2023.06 2025.06

*Supported CPU families:* AMD Intel Arm

*Supported GPU families:* NVIDIA

## GROMACS

GROMACS is a versatile package to perform molecular dynamics, i.e. simulate the Newtonian equations of motion for systems with hundreds to millions of particles.

This is a GPU enabled build, containing both MPI and threadMPI binaries.

It also contains the gmxapi extension for the single precision MPI build.

homepage: https://www.gromacs.org

### Available installations

| GROMACS version | Supported CPU targets | Supported GPU targets | EESSI version | Module |
|---|---|---|---|---|
| 2024.1 | generic : aarch64, x86_64<br>Arm : a64fx, neoverse_n1, neoverse_v1, nvidia/grace<br>AMD : zen2, zen3, zen4<br>Intel : haswell, skylake_avx512, sapphirerapids, icelake, cascadelake | (none) | 2023.06 | GROMACS/2024.1-foss-2023b |
| 2024.4 | generic : aarch64, x86_64<br>Arm : neoverse_n1, neoverse_v1, nvidia/grace<br>AMD : zen2, zen3, zen4<br>Intel : haswell, skylake_avx512, sapphirerapids, icelake, cascadelake | NVIDIA : cc70, cc80, cc90 | 2025.06 | GROMACS/2024.4-foss-2023b-CUDA-12.4.0 |
| 2024.4 | generic : aarch64, x86_64<br>Arm : a64fx, neoverse_n1, neoverse_v1, nvidia/grace<br>AMD : zen2, zen3, zen4<br>Intel : haswell, skylake_avx512, sapphirerapids, icelake, cascadelake | (none) | 2025.06 | GROMACS/2024.4-foss-2023b |

### Extensions

Overview of extensions included in GROMACS installations

gmxapi

| gmxapi version | GROMACS modules that include it |
|---|---|
| 0.4.2 | GROMACS/2024.4-foss-2023b-CUDA-12.4.0<br>GROMACS/2024.3-foss-2023b<br>GROMACS/2024.3-foss-2023b-PLUMED-2.9.2<br>GROMACS/2024.4-foss-2023b<br>GROMACS/2025.2-foss-2025a |
| 0.5.0 | GROMACS/2024.1-foss-2023b |

# EESSI documentation

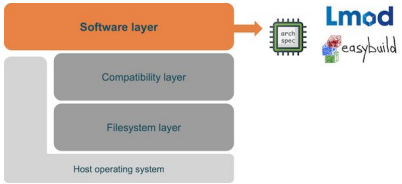
eessi.io/docs - GitHub repo github.com/EESSI/docs

Highlights:

- EESSI Events & Training
  - Added EESSI Happy Hour ( + added links to recordings)
- Blog posts (https://www.eessi.io/docs/blog/)
  - EESSI hackathon at EMBL
  - EESSI at FOSDEM ‘26
  - 2nd post on using Spack on top of EESSI
- GPU instructions for 2025.06 added (and updated in general)
- CVMFS tutorial exposed at https://www.eessi.io/docs/tutorial
  - Still hard to find though
- Plus other minor changes to the documentation (software overview, corrections,...)

[Lara, Pedro, Helena, Bob, Xin, Kenneth, Caspar…]

# EESSI test suite




eessi.io/docs/test-suite - github.com/EESSI/test-suite

ReFrame

## Highlights

- Current release: v0.9.0
- Number of fixes/improvements to existing suite:
  - fix process binding for mpirun with OpenMPI (PR #305)
  - Small fix so that you can load more than 1 local module (PR #323)
  - Number of other fixes: PR #301, PR #302, PR #312, PR #315, PR #316, PR #318

[Caspar, Sam, Satish, Kenneth, Lara]

# Status update of EESSI on EuroHPC systems

- EESSI is currently available on 7 out 9 current EuroHPC supercomputers: Vega, Karolina, Deucalion, MareNostrum 5, Discoverer, MeluXina, Leonardo*
- Work-in-progress to make EESSI available on other 2: LUMI, JUPITER
- EESSI is expected to be available on all 9 EuroHPC systems by end of March 2026, thanks to EuroHPC Federation Platform (EFP)
  - Upcoming EuroHPC AI Factories will also be required to integrate with EFP
- 1st release of EFP is due end of March 2026
  - Will include EESSI as base of Federated Software Catalog component (in beta)
  - More info on EFP: https://my-eurohpc.eu

[Kenneth, Alan, Lara]

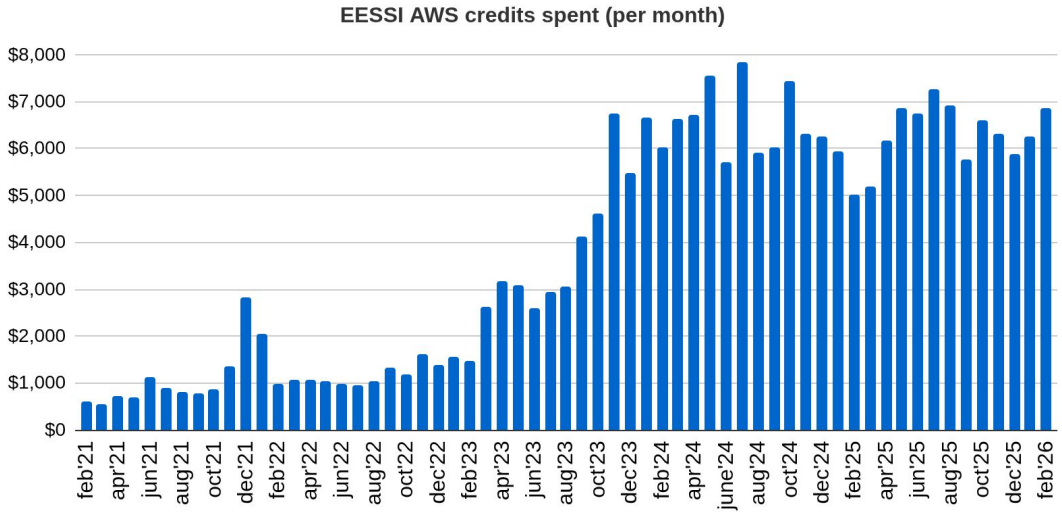
# Improved NVIDIA GPU support

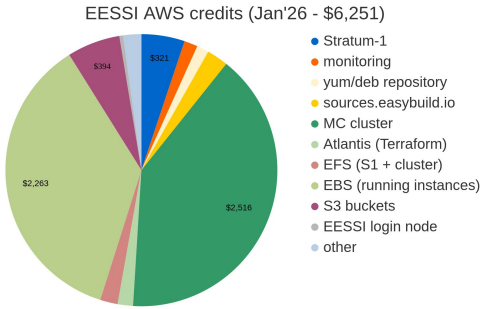
- CUDA builds for 2025.06 are now available (PR #1351)
  - CUDA 12.6.0 + cuDNN 9.5.0.50,  CUDA 12.8.0 + cuDNN 9.10.1.4
  - Working on CUDA 12.9.1 + cuDNN 9.15.0.57 (PR #1410)
  - Unsupported compute capabilities have a dummy module file printing an error
- New version of the host libraries linking script
  - `/cvmfs/software.eessi.io/versions/2025.06/scripts/gpu_support/nvidia/link_nvidia_host_libraries.sh`
  - Makes use of the new variant symlinks to inject the host libraries:
    i. EESSI_NVIDIA_OVERRIDE_DEFAULT: version-agnostic path
    ii. EESSI_202506_NVIDIA_OVERRIDE: version-specific path
  - The default points to `/dev/null`, so one of them has to be set!
  - See https://www.eessi.io/docs/site_specific_config/gpu/#nvidia_drivers

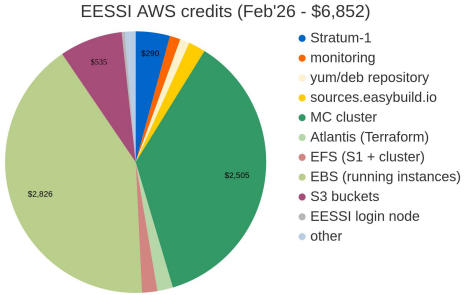
[Caspar, Bob]

# Sponsored AWS credits

- ~$6,251 + $6,852 “spent” in Jan'26 + Feb'26 on Stratum-1 servers, monitoring, demos, sources.easybuild.io, Slurm clusters (build bot)
- **Bulk of consumed credits due to EESSI build-and-deploy bot**
- **New build cluster in Barcelona (eu-south) for access to Zen 5**
- Regular sync meetings with AWS
- Received a new (big) batch of credits for 2026 🎉




[Kenneth, Bob, Alan]

# EESSI CLI tool (WIP)

- https://github.com/EESSI/eessi-cli , `pip install eessi`
- Working subcommands:
  - `eessi init` (needs more work to be really useful)
  - `eessi check` to check on CernVM-FS configuration (more checks coming)
  - `eessi shell` to start a subshell in which a particular EESSI version is available
- WIP: `eessi install` to install & configure CernVM-FS to get access to EESSI, ee https://github.com/EESSI/eessi-cli/pull/16
- Ideas for additional subcommands & features: https://hackmd.io/NASotth6R9aoTQTQx9VqLg?view
- See it in action: https://eessi.io/docs/blog/2026/02/04/EESSI-at-FOSDEM-26/#eessi-cli-tool

[Kenneth]

# EESSI Governance - (interim) Steering Committee

- **Governance for EESSI has been formalized in the EESSI documentation**
  - https://eessi.io/docs/governance
  - Feedback is still very welcome! A procedure is in place to consider (and vote on) changes
- Current members of EESSI Steering Committee (ESC):
  Kenneth (UGent) - chair, Bob (UGroningen), Caspar (SURF), Thomas (UBergen), Helena (Do IT Now)
- Most recent ESC meeting was held on 19 Feb 2026
- Agenda items of last ESC meeting:
  - Short-term roadmap for EESSI
  - Identifying and eliminating Single-Points-of-Failure (SPOFs)
  - Briefly: EuroHPC funding, EESSI “releases”, …

[Kenneth]

# EESSI project @ EMBL hackathon

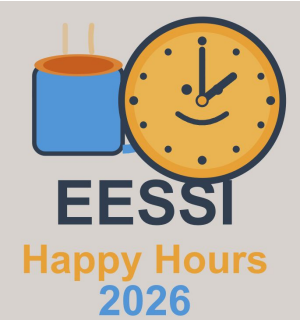
- https://grp-bio-it.embl-community.io/hackathons/#projects
- 3C-CoDash EMBL Event 2026, 28-30 Jan 2026 @ EMBL, Heidelberg and online
- Quote from project description: *"EESSI will be the primary and default source of software in the next iteration of the EMBL HPC cluster."*
- Blog post: https://www.eessi.io/docs/blog/2026/01/31/easy-EESSI-at-EMBL-CoDash/
- Small group (5) focusing on this particular project during the 3-day hackathon
  - Team members **submitted easyconfigs for evaluation by EESSI**
  - Work began on **extending an internal EMBL GitLab CI pipeline** so it can build on top of EESSI
  - Experience generated **first contributions** toward EESSI and laid groundwork for future work before the next hackathon.

[Kenneth, Renato @ EMBL]

# EESSI Happy Hour sessions ⏰ **Mondays 14:00-15:00** CE(S)T

**EESSI Happy hour sessions 2026:**

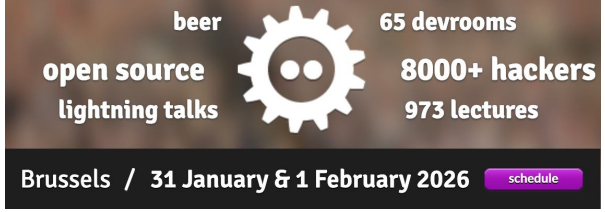
- Started Monday January 12 with EESSI Roadmap 2026
- 8 sessions so far in 2026
- Focusing on ways to get access to EESSI
  - EESSI on macOS: Bringing HPC Software to Your Mac
  - Have You Met the EESSI Command-Line Interface (CLI)?
  - EESSI Offline: Portable CVMFS Repositories
  - EESSI Without Native CVMFS? Meet cvmfsexec!
  - Revisit eessi_container.sh
  - Meet EESSI: Build Once. Run Everywhere
- Upcoming EESSI Happy hour sessions
  - Yet another way to run EESSI: Docker Extensions
  - From x86 to RISC-V: Expanding EESSI's Reach




https://eessi.io/docs/training-events/happy-hours-sessions/

[Richard]

# EESSI @ FOSDEM'26




- First weekend of February in Brussels
- FOSDEM is an open source software event, free to attend (no cost, *no registration*)
- Recordings available for talks related to EESSI, see

  https://eessi.io/docs/blog/2026/02/04/EESSI-at-FOSDEM-26

[Kenneth,Thomas, Helena, Lara, …]

# Webinar series on EuroHPC Federation Platform

- https://my-eurohpc.eu/training/
- Via Zoom, open webinars, **registration required**
- Recordings available for 3 webinars: Introduction to EFP, **Federated Software Catalog (based on EESSI)**, EFP Interactive
- More webinars to come:
  - Wed 29 Apr'26 14:00 CEST: EFP Authentication & Authorization Infrastructure (AAI)
  - Thu 21 May'26 14:00 CEST: EFP Workflows

[Kenneth, Alan]

# 11th EasyBuild User Meeting

https://easybuild.io/eum26

- Tue-Thu 21-23 April 2026 @ Guimarães, Portugal (home of Deucalion)
- **3-day event, day 3 will focus again on EESSI**
- Initial agenda is available, more details to come
- Suggestions/proposal for talks are still welcome
- Registration is open (40 seats max. for in-person attendance)

[Kenneth]

# ISC’26 conference

- https://isc-hpc.com
- 22-26 June 2026 @ Hamburg, Germany
- Proposal for half-day tutorial on EESSI accepted 🚀🚀🚀 (Mon 22 June 2026)
- BoF proposals submitted on EESSI & EuroHPC Federation Platform (EFP)
- Talks/demos on EESSI at various booths at ISC exhibit (to be confirmed)
- Most likely also some kind of social meetup… 🍻

[Kenneth]

# Upcoming EESSI webinars

- **Save the date!** Each webinar will start at 14:00 CEST
  - Mon 27 Apr'26: *Introduction to EESSI* - Helena
  - Mon 4 May: *Building software on top + adding software to EESSI* - Kenneth + Lara
  - Mon 11 May: *Using EESSI for Continuous Integration (CI)* - Alan
  - Mon 18 May: *Introduction to CernVM-FS* - Valentin Volkl
- More info (incl. registration) will be available soon via https://eessi.io/docs/training-events
- A separate webinar series is being planned for EasyBuild

# EESSI mailing list eessi@list.rug.nl

- Low-frequency mailing list
  - Announcing bi-monthly meetings
  - Important updates and announcements
- To (un)subscribe, send an email to listserv@list.rug.nl with the following body:
  - Subscribe:

    ```
    SUBSCRIBE EESSI Firstname Lastname
    ```

  - Unsubscribe:

    ```
    UNSUBSCRIBE EESSI
    ```

# Next EESSI update meetings

- EESSI update meetings are **bi-monthly** (odd months)
- First Thursday of the month at 14:00 CE(S)T, only in odd months
  - iCalendar URL for calendar integration
- Next meetings:
  - Thu 7 May 2026 14:00 CEST (12:00 UTC)
  - Thu 2 July 2026 14:00 CEST (12:00 UTC)
  - Thu 3 Sept 2026 14:00 CEST (12:00 UTC)